F79SP Session 18: Stationary and Limiting distributions of Markov processes

1 Theory of stationary and limiting distributions

Recall the notion of irreducibility and aperiodicity from Markov chains.

- Similarly, we call a Markov process **irreducible** if it state space forms a single closed class.
- A Markov process **does not have periodicity issue** inside any closed class, since $i \to j$ implies $p_{ij}(t) > 0$ for **all** $t > 0$ (though its embedded Markov chain may be periodic).

Theorem 1.1. *Assume that the state space $\mathcal{S}$ is a single closed class for the Markov process. Suppose that the distribution $\boldsymbol{\pi}$ is **stationary**. Then $\pi_j > 0$ for all $j \in S$.*

Proof. Since $\sum_{i \in S} \pi_i = 1$ we have $\pi_i > 0$ for *some* $i \in S$. Then, since S is a single class (all states intercommunicate), for any other $j \in S$ and for any $t > 0$, $p_{ij}(t) > 0$, and so, from $\sum_{i \in S} \pi_i p_{ij}(t) = \pi_j$, we have $\pi_j > 0$. □

Theorem 1.2. *Consider a Markov process. Suppose that S is a **single closed class**. Then* either

1. *there is a **unique stationary distribution $\boldsymbol{\pi}$** and then*

$$\lim_{t \to \infty} p_{ij}(t) = \pi_j, \qquad \text{for all } i, j \in S;$$

*in this case S is **positive recurrent** with respect to $\{X(t)\}_{t \geq 0}$;*

or

2. *there is **no stationary distribution** and then*

$$\lim_{t \to \infty} p_{ij}(t) = 0, \qquad \text{for all } i, j \in S;$$

*in this case S is **null recurrent** or **transient** with respect to $\{X(t)\}_{t \geq 0}$.*

If S is finite then the case 1 necessarily holds.

Result:

- For an **irreducible** Markov process with **finite state space**, there exists a **unique stationary distribution**, and this stationary distribution is **equal to the limiting distribution**.
- For the embedded Markov chain, if in addition, it is aperiodic, then it will have a unique stationary distribution, and this stationary distribution is equal to the limiting distribution.
- **Question**: When does the stationary distributions of a Markov process and its embedded MC coincide?
- **Answer**: This will happen if and only if q_{ii} **is the same for all** $i \in S$ **(a single closed class)**. In this case, the time between jumps will always be $Exp(q_{ii})$ distributed.

2 Reducible Markov processes

We consider Markov processes with more than one closed classes.

Example 2.1. *Consider a Markov process* $\{X(t)\}_{t\geq 0}$ *with state space* $S = \{1, 2, 3, 4, 5\}$ *and transition rate matrix:*

$$Q = \begin{pmatrix} -2 & 0 & 1 & 0 & 1 \\ 0 & -5 & 0 & 5 & 0 \\ 2 & 3 & -6 & 0 & 1 \\ 0 & 2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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1. *Classify the state space of MP.*
2. *Find general form of stationary distribution of MP.*
3. *Find limiting distribution of MP for* $X(0) = 2$.
4. *Find limiting distribution of MP for* $X(0) = 3$.
5. *Are stationary distributions of MP and jumping chain (embedded MC) the same?*
6. *Find stationary distribution of jumping chain.*

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